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RESEARCH ARTICLE

A Study on Rephrasing Large Earthquakes via the Least Action Principle and Evolving Machine Learning

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ABSTRACT Understanding large earthquakes (EQs) has been a critical endeavor. Recent advancements in machine learning (ML) help this endeavor, yet they face formidable performance ceilings due to our technical limitations, restricted access to seismogenic fault zones, and the lack of large datasets for ML training. This paper introduces a novel approach by conceptualizing the EQ-bearing lithosphere as a chaotic system and investigating whether the least action principle (LAP) from Hamiltonian mechanics can illuminate the potential predictability of large EQ initiation. Leveraging an abstract space defined by time and rupture derivatives, this paper establishes a potential causal pathway between novel LAP-based features and large EQs. An evolving ML framework utilizing parallel evolution strategies was developed to identify hidden rules derived from LAP. The model was trained on data from all EQs in the western U.S. over the past 40 years. Feasibility tests on historical large EQs show promising performance compared to state-of-the-art forecasting methods. While practical applicability requires deeper investigation, these findings suggest a new research direction for large EQ prediction that merges classical physics and evolving ML.

INDEX TERMS Artificial intelligence for large earthquakes, evolving machine learning algorithms, least action principle for chaotic earthquakes, parallel evolution strategies, scientific machine learning for geophysics.

I. INTRODUCTION

In broad science and engineering, advanced artificial intelligence (AI) and machine learning (ML) methods gradually outperform the existing computational prediction methods that are based on the sophisticated domain knowledge and underlying first principles, e.g., global weather [1], holistic Earth system [2], biomolecular interactions [3] to name a few. Pushing the boundary with AI can also be found in geoscience and seismogenesis [4], [5], [6], [7], [8], [9], [10], [11], [12]. Also, Table 1 summarizes the recent AI and machine learning approaches for earthquake prediction and forecasting.

Motivation: AI-empowered approaches for large earthquakes (EQs), however, face the formidable performance

ceiling. Admittedly, we cannot predict short-term EQs in terms of locations, magnitudes, and timings [13], [14], [15], [16]. This obstacle is attributed to our fundamental inability, i.e., the limited access to seismogenic fault zones and the lack of large data which are vital for training AI. However, persistent efforts are underway – e.g., AI-driven seismological data enhancements [9], [13], transparent AI-friendly data transformations for large EQs [17], [18], [19], and the pursuit of precursory for EQs [20].

Contribution: This paper's contributions to the body of large EQ research include (i) a novel adoption of the least action principle to holistically describe large EQs, (ii) a search of explainable rules of the large EQs' magnitudes and locations within a short time window, and (iii) evolving ML for autonomous rule searching with 40-years long EQ data of the western U.S. In particular, this paper seeks to

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add a new dimension to large EQ research by answering the following question – “at a certain time window $[t_a, t_b]$, how does the Earth lithosphere select a certain set of fault zones to rupture? Is it purely random, being impossible to predict. Or is it a chaotic process, being subject to a deeper and more fundamental (hopefully succinct) hidden rule?” This paper accepts the perspective of a chaotic process [21] in which earthquakes are formidably complex and sensitive to the initial condition (so-called “butterfly effect”), but they are predictable. Responses to this fundamental question may vary, holding their own values. Some focus on individual fault mechanisms [22], [23], [24] while others highlight statistical and thus practically meaningful behaviors [25], [26]. This paper regards earthquakes as a chaotic process and seeks to find a succinct yet fundamental rule from the hundreds-year-old principle, the “Hamilton’s principle” (also called the least action principle (LAP)) [27]: $\delta \int_{t_a}^{t_b} (T - V)dt = 0$ where T and V are the kinetic and potential energy (PE) of the entire system, respectively. In essence, this paper hypothesizes that the lithosphere system as a whole chooses a certain set of EQs within any time window $[t_a, t_b]$ by following the LAP and that such EQ generation patterns (or rules) can be unraveled by inheriting Hamiltonian mechanics in an abstract space of time and ruptures. Fig. 1(A) illustrates basic assumption of this paper about EQ cycles, and Fig. 1(B) summarizes the overall architecture of the proposed AI framework.

Organization: This paper is organized to present the derivations of LAP applied to large EQs (Section II-A). Then, Section III demonstrates how evolving AI methods can facilitate the pursuit of LAP-based large EQ generation rules through pseudo-prospective predictions, comparing the proposed AI framework with the state-of-the-art EQ forecasting method, UCERF-ETAS (Uniform California Earthquake Rupture Forecast with Epidemic Type Aftershock Sequence). Finally, Section IV presents key insights, limitations, and directions for future work.

II. FUSION OF PHYSICS AND MACHINE LEARNING

A. LEAST ACTION PRINCIPLE FOR LARGE EARTHQUAKES

$$\delta \int_{t_a}^{t_b} (T(\dot{R}_1, \dots, \dot{R}_n) - V(R_1, \dots, R_n))dt = 0 \quad (1)$$

where we regard the kinetic energy T of the entire system as a function of the j th rupture zone’s rupture velocity (i.e., $\dot{R}_j = dR_j/dt$) and PE (V) of the rupture $R_j(t)$. In this formulation, there can be as many active rupture zones as the total reference volumes. For brevity, the fault’s hypocenter location ξ is omitted in the following. Here, each rupture zone is considered as a Lagrange’s generalized coordinate. It should be noted that the “rupture” R is not a fixed or pre-determined term, but it is a time-varying quantity, meaning the total length of a fault zone that slips during an earthquake event. The generic solution to the LAP in the system of n generalized coordinates (i.e., rupture zones) is

TABLE 1. Recent AI/ML approaches for EQ prediction and research.

Year [Ref]	Recommended AI/ML	Main Targets and Findings
2025 [49]	Fully Convolutional Networks	EQs with $Mw \geq 3.0, 4.0$, and 5.0 like ETAS model
2025 [27]	RNN-LSTM*	EQ timing, location, and magnitude
2025 [23]	Multimodal fusion deep neural networks	Forecasting of 50-years EQs in China and Transfer to EQs in USA
2025 [26]	Self-supervised automatic speech recognition	Contemporaneous EQ fault displacement
2024 [14]	Reinforcement Learning for large EQs	Explainable rules of short-term prediction of large EQs
2024 [47]	Random Forest	30-days ahead maximum potential EQ magnitude
2023 [48]	Seasonal Autoregressive Integrated Moving Average	short-time predictions of small to modest EQs
2023 [13]	AI-friendly new feature generation with Fourier Transform	short-time predictions of large EQs in the western USA
2023 [36]	Decision Trees after comparative study with recent ML methods	EQ magnitude, trend and occurrence
2023 [31]	Comprehensive review of ML methods for EQs	Broad EQ seismology
2022 [46]	Deep Learning-based noise suppression	Urban earthquakes monitoring
2022 [11]	Gauss curvature-based new feature for ML	Short-term prediction of large EQs

* RNN: Recurrent Neural Networks; LSTM: Long Short-Term Memory.

the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial(T - V)}{\partial \dot{R}_j} \right) - \frac{\partial(T - V)}{\partial R_j} = 0, \quad (j = 1, \dots, n) \quad (2)$$

Since we assume $T = T(\dot{R})$ and $V = V(R)$, these equations are simplified as $\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{R}_j} \right) = -\frac{\partial V}{\partial R_j}$, ($j = 1, \dots, n$). In general, T can have an arbitrary form of the rupture velocity $\dot{R}$, but geophysical observations offer some physically meaningful and exclusive conditions about possible T . This paper proposes $T(\dot{R}) = c(\dot{R})^3$ with c , a constant, for which physical rationales are presented in Discussion section.

As proven in Appendix, the Euler-Lagrange equations to the LAP for EQ generation boil down to the following equations

$$\dot{R}_j^2 \ddot{R}_j = \frac{A_{max}}{3cP} \sum_{k=1}^{n_{st}} (-1)^k \cos \left(\frac{2\pi k}{P} (t + \phi) \right), \quad (j = 1, \dots, n). \quad (3)$$

where A_{max} is the amplitude; P is the period; $\phi(t; R(t, \xi_j))$ is the phase of the sawtooth wave at time. n_{st} is the total number of base trigonometric functions of the sawtoothwave ($n_{st} = 100$ is used). By solving these partial differential equations, we can obtain the LAP-driven EQ rupture expression (see Appendix) and thus obtain a direct relationship between the rupture dynamics and PE variation:

$$\frac{\partial V}{\partial R_j} = -6c\dot{R}_j \ddot{R}_j, \quad (4)$$

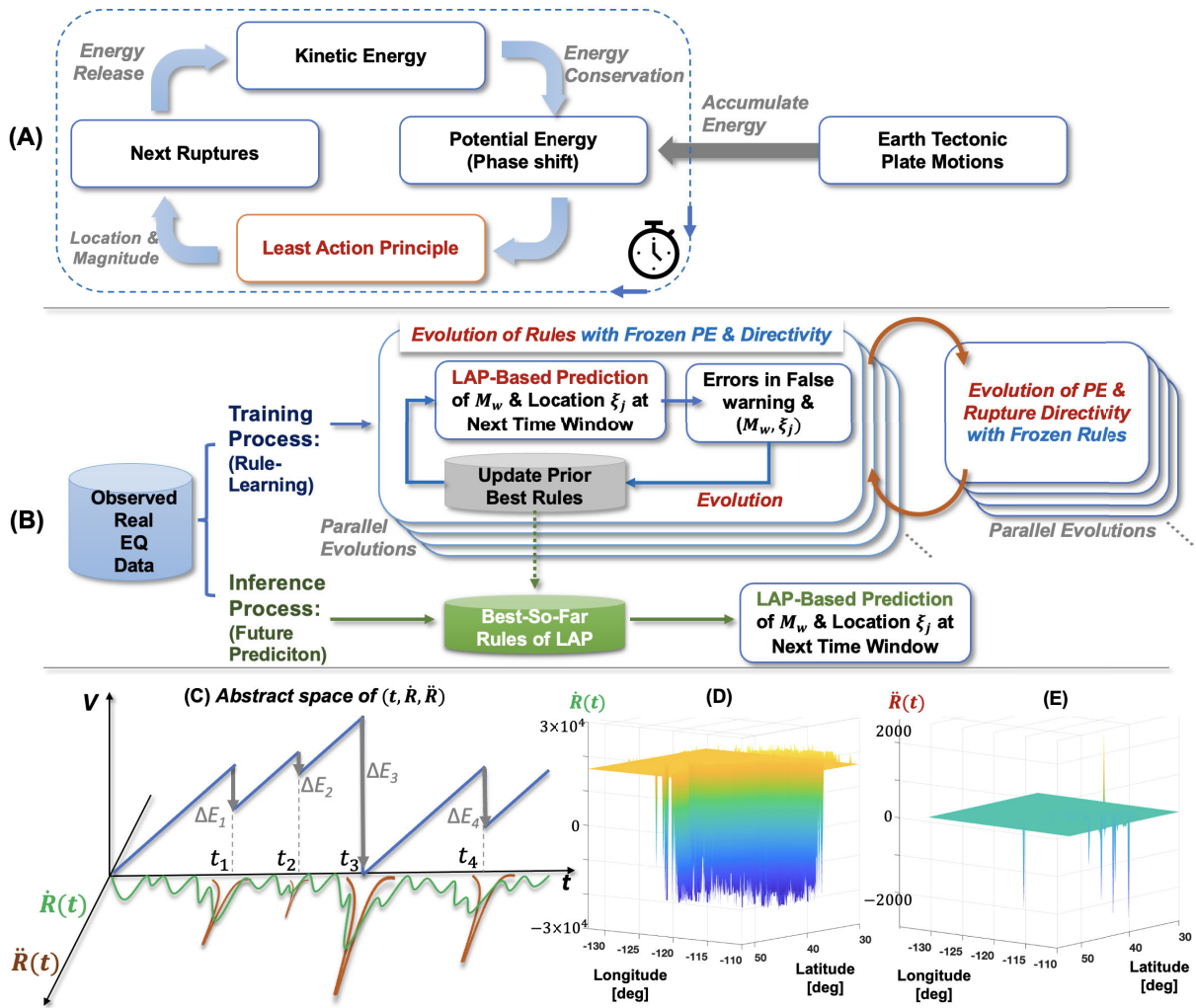


FIGURE 1. The Least Action Principle (LAP) in the Cycle of Earthquake (EQ) Generation and AI-Driven Evolution of Rules. (A) Overall Role of LAP in EQ Cycles: This paper mainly hypothesizes that within any time interval the earth lithosphere satisfies LAP to determine the locations and magnitudes of next ruptures, thus releasing kinetic energy. By the energy conservation, the PE function is shifted to balance the loss of energy by EQs. This constitutes an incessant cycle (dashed box) being fueled by the tectonic plate motions of the Earth. (B) Flow of Training and Inference: The training process employs a parallel Evolution Strategy (ES) to coherently evolve EQ- and LAP-based rules while keeping V and directivity fixed. Conversely, V and directivity are evolved while the rules remain frozen. This two-stage cyclic evolution runs across parallel cores (e.g., >100 CPUs). During inference, the best-so-far LAP-based rules are applied to predict EQs for the future time window. (C) Abstract Space of Time and Derivatives of Rupture Length $R(t)$ of LAP. Example plots of the first derivative of rupture, i.e., rupture velocity (D) and the second derivative of the rupture, rupture acceleration (E). These example plots are drawn from the Ridgecrest EQ ($M_w = 7.1$, 2019, July 6).

Full derivations and expressions of $R(t)$, $\dot{R}$, and $\ddot{R}$ are presented in Appendix. Fig. 1(C) shows some example of the rupture derivatives and abstract space of this paper.

B. ENERGY, MOMENT MAGNITUDE, AND RUPTURE LENGTH

The heuristic relationship between moment magnitude (M) and energy (E) has been suggested by many researchers (e.g., [36]), and generally they take a logarithmic form as

$$M = \alpha_1 + \beta_1 \log_{10} E \quad (5)$$

On the other hand, the heuristic relationship between the subsurface rupture length (R) and moment magnitude was

often suggested by another logarithmic form (e.g., [37]):

$$M = \alpha_2 + \beta_2 \log_{10} R \quad (6)$$

From these two logarithmic relations we can simply derive the relation of energy and rupture range as

$$R = 10^{(\alpha_1 - \alpha_2)/\beta_2} E^{\beta_1/\beta_2} \quad (7)$$

For instance, when $\alpha_1 = -2.93$, $\beta_1 = 0.67$, $\alpha_2 = 3.76$, and $\beta_2 = 2.0$, we have $R(t, \xi_j) = 10^{-3.345} E(\xi_j)^{1/3}$ where the energy $E(\xi_j)$ is regarded as the released energy from EQ at ξ_j . Inversely, we can have $E(\xi_j) = (10^{3.345} R(t, \xi_j))^3$. It should be noted that all the coefficients used in the aforementioned rules will be learned and evolved by the machine learning (e.g., evolution strategy (ES)).

C. POTENTIAL ENERGY WITH SAWTOOTH WAVE FUNCTION

As the major premise, this paper assumes a mathematical form of the PE of the lithosphere that can effectively describe PE being constantly charged by the incessant internal movement of the earth. PE of each reference volume (i.e., the discretized volume of lithosphere with a position vector $\xi_j \in \mathbb{R}^3$) is denoted by $V(t; \xi_j)$ at time t , and PE is implicitly related to the rupture length $R(t, \xi_j)$ via the phase shift ϕ as:

$$V := A_{max} \left(\frac{1}{2} - \frac{1}{\pi} \sum_{k=1}^{n_{st}} \frac{(-1)^k}{k} \sin \left[\frac{2\pi k}{P(\xi_j)} (t + \phi(t; \xi_j)) \right] \right) \quad (8)$$

$$\phi(t; \xi_j) = \sum_{\tau=0}^{t-1} \Delta\phi_{small}(\tau; R(\tau, \xi_j)) \text{ or } \phi(t; \xi_j) \leftarrow \Delta\phi_{large} \quad (9)$$

where $A_{max}(\xi_j)$ is the amplitude meaning the maximum PE of the j -th reference volume. $P(\xi_j)$ is the period of the sawtooth PE function and assumed position dependent. $\phi(t; R(t, \xi_j))$ is the phase of the sawtooth wave at time t that is slightly adjusted by all the past small rupture lengths $R(\tau)$, $\tau < t$ (10) or abruptly reset by a large shift $\Delta\phi_{large}$ of (II-C). Fig. 2 shows examples of two sawtooth waves with varying n_{st} . This paper assumes $n_{st} = 100$ to ensure an sufficiently smooth rise and sudden drop after passing the peaks. Still, there is no restriction to the choice of higher-order sawtooth function. It is assumed that the amplitude A_{max} and the period P of PE are location-dependent and should be learned by AI. The phase shift ϕ is the function of time and location to be adjusted by AI in such a way that the energy conservation is satisfied. During the training, the phase shift of each location is calculated and updated according to the energy balance (i.e., compensating the released kinetic energy (KE) of EQs). Fig. 3 illustrates how phase shift is used to satisfy the conservation of energy. Fig. 3(A) explains how the observed KE of an small EQ is balanced by shifting the PE curve whereas Fig. 3(B) shows how a large EQ is matched by resetting the PE curve to zero energy state. When a small EQ's rupture length does not exceed the dimensions of one reference volume, a slight phase shift is calculated by balancing increments of PE and KE as illustrated in Fig. 3A.

$$\Delta\phi_{small}(t; R(t, \xi_j)) = -T_K^{(dist)}(R(t, \xi_j))P(\xi_j)/A_{max}(\xi_j) \quad (10)$$

where $T_K^{(dist)}(R(t, \xi_j), \xi_j)$ is the distributed kinetic energy that is based on the rupture length $R(t, \xi_j)$ at location ξ_j (details are presented in *Distribution of Kinetic Energy*). When a large EQ's rupture spans multiple reference volumes, all the associated reference volumes experience sudden drop of PE as depicted in Figs. 3(B). Such a large phase shift is calculated by $\phi(t; \xi_j) \leftarrow \Delta\phi_{large}$ where

$$\Delta\phi_{large} = \left\lceil \left(\frac{t_{now} - 0.5P}{P} \right) P + \frac{P}{2} - t_{now} \right\rceil + 10^{-3}P$$

The bottom line of this learning process is that we are unaware of the "true" PE function and that the observations of EQs can offer hints at phase of PE at the present time. As investigated by [38], the alignment of stress/strain level among connected or nearby seismogenic zones is critical for determining EQ arrest or cascading failure. Therefore, this paper's learning process seeks to align phase shifts across and KE distributions more than one reference volume when large EQs occur. In this regard, rupture directivity is mainly used and it is also learned by observed EQs data (details in later section).

D. RUPTURE ZONE SUBDOMAIN AND DATA-DRIVEN RUPTURE DIRECTIVITY

This paper assumes two types of subdomain Ω_j as

$$[\text{Spherical subdomain } \Omega_j := \{\xi_i | \forall \xi_i \in V, \|\xi_i - \xi_j\|_2^2 = R^2\}. \quad (11)$$

$$[\text{Seismogenic rupture zone } \Omega_j := \{\xi_i | \forall \xi_i \in V\}, \quad (12)$$

2D surface with dimensions $R \times W$.

The directivities of the seismogenic rupture zone Ω_j can be learned from the observed EQs of the USGS database [39]. To enable data-driven learning and evolution of the rupture directivity, this framework assumes that we are unaware of directions of all faults of the entire domain and seeks to identify them gradually. Although the direction can be arbitrary, for computational efficiency, this paper assumes in total 12 directions of potential rupture zones (Fig. 4A). As illustrated in Fig. 4B, each hypocenter is regarded as the center location, and directional cosines are calculated between the center and nearby EQ hypocenters. These angles are accumulated and assigned into 12 categories according to the scheme illustrated by Fig. 4A. This paper assumes all ruptures lie on XY planes (i.e. horizontal plane) at each of 5 different depths (i.e., at -2.5 km, 2.5 km, 7.5 km, 12.5 km, and 17.5 km). At each depth, strike-slip motions (dip fixed) are assumed on the plane, and thus the 12-interval binning scheme used herein can be understood as the group of strikes of rupture planes. The relative frequency of directional events (among 12 categories) at each location forms the probability density function (PDF) of the directivity (e.g., Fig. 4D-E). The maximum-probability directivity will be used as the probable rupture direction at the location. For instance, 40-years entire EQ catalog data are used in the initial training and the AI framework can obtain the initial map of maximum probability rupture directions at 5 depths. Fig. 4C shows one of such plots at depth 12.5 km. Similar colors indicate common rupture directivities. For Ω_j , we can consider two scenarios. On the one hand, we search for a sphere of reference volumes of which center is at ξ_j (Fig. 4F). The sphere is regarded to contain possible ruptures associated with the EQ. Since a rupture can propagate as much as R in any direction from ξ_j , the radius of R would define a conservatively large spherical domain. Physically, this sphere is assumed to lose its stored PE via the released kinetic energy

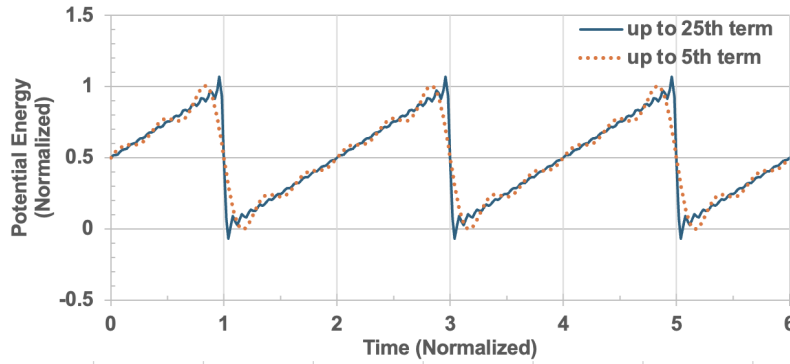


FIGURE 2. Sawtooth Wave Function for Potential Energy: Two waves are shown up to 5-th term and 25-th term of the sawtooth function. The period 2 is used for both cases, and PE in vertical axis is normalized by the maximum amplitude to make range [0,1].

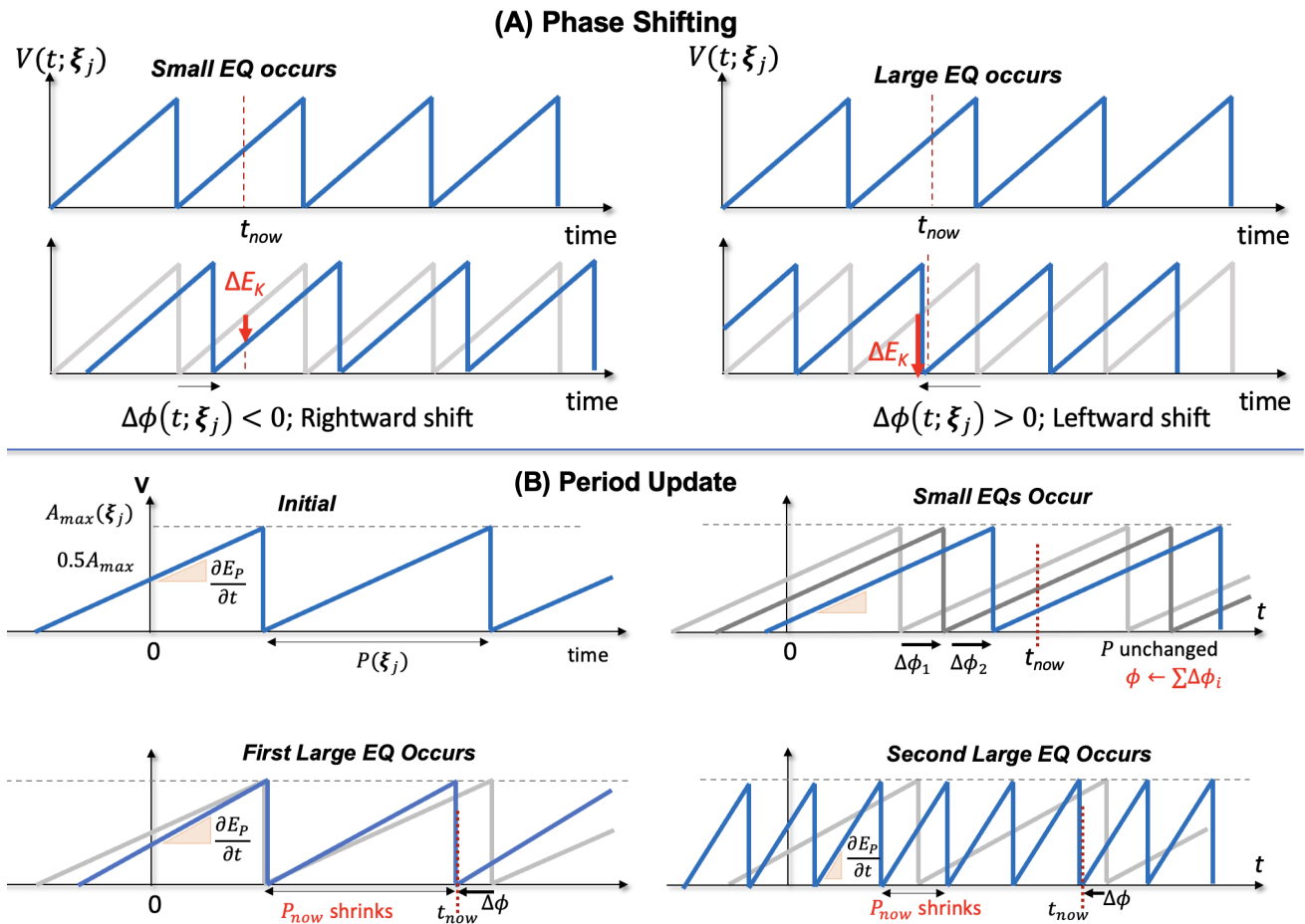


FIGURE 3. Evolution Processes of the Potential Energy Function: (A) Phase Shifting Process: When a small to moderate magnitude EQ occurs, the PE decreases by the amount of kinetic energy ΔE_K , which is described by a negative phase shift of the sawtooth wave. When a large magnitude EQ occurs, all the stored PE is assumed to be released, and it is described by the sudden positive phase shift. (B) Period Update Process: Initial PE with (16) and $P = 365,000$ days at time t . When small EQs occur, only phase shifts are calculated and accumulated while no adjustment on P is done. When a large EQ occurs for the first time, the period shrinks whereas next large EQ occurs, additional shrink of period is done.

emitted from EQs. We are not certain about the decisive parameters used in (6) and (7) and the precise dimension of the sphere, but the energy conservation is imposed therein. On the other hand, we can restrict the subdomain Ω_j as a

narrow seismogenic rupture zone with length R and width W (Fig. 4G). Compared to the spherical domain, this narrow rupture zone concept substantially reduces the reference volumes that involve in active EQ and thus the released

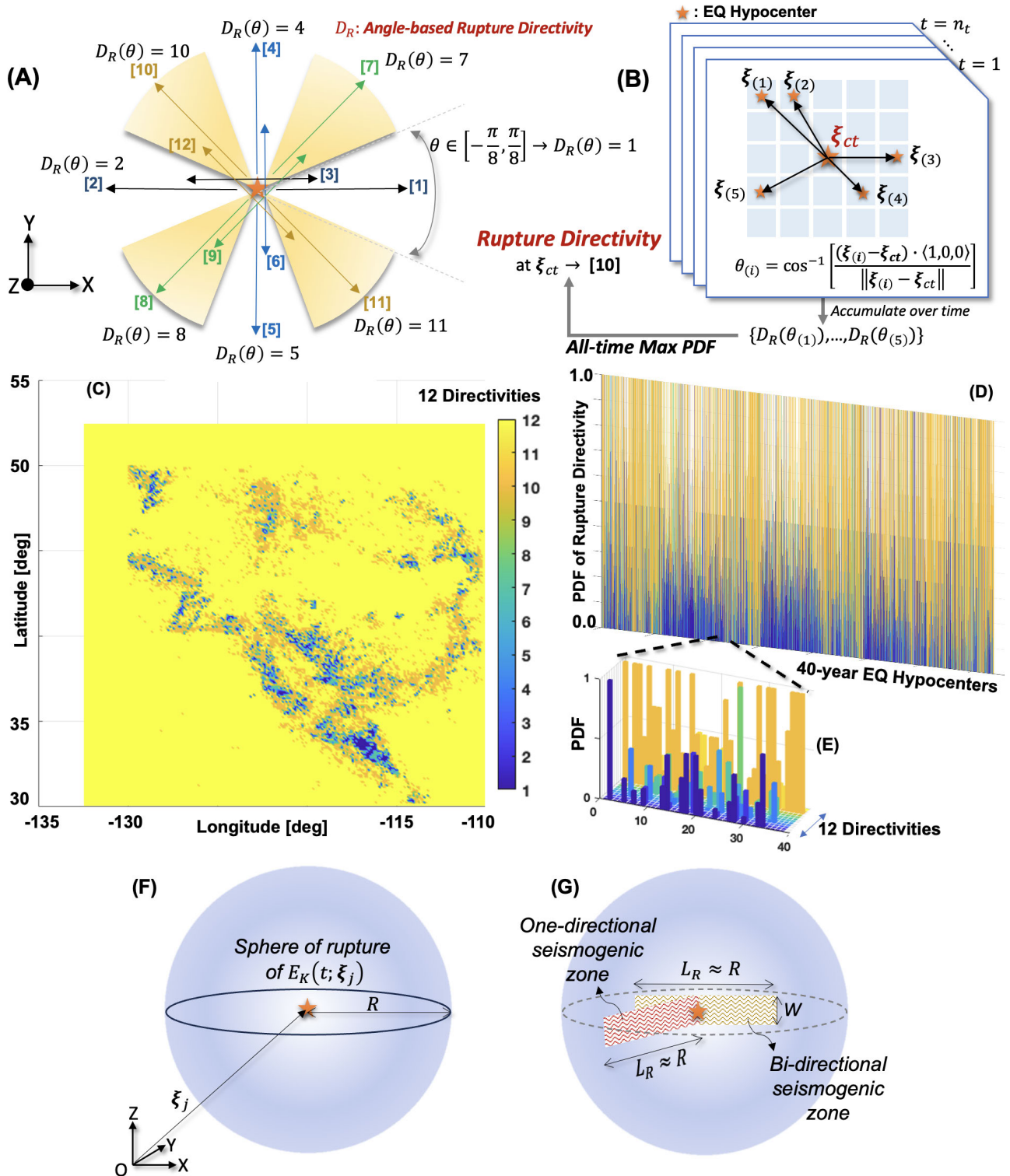


FIGURE 4. Rupture Directivity Determination Algorithm: (A) In total, 12 categories (each spans $\pi/4$ radian) of possible rupture directivity at each reference volume are assumed. Unidirectional (e.g., [1]) or bi-directional (e.g., [3]) rupture directivities are included; (B) Directional cosines between a center hypocenter and nearby hypocenters define a set of angle-based rupture directivity. Over time, event frequency-based probability density function (PDF) is constructed at the center hypocenter. The maximum-probability rupture directivity is assigned to that location; (C) Maximum-probability rupture directivity map obtained from the proposed algorithm with the entire 40-year EQ catalog; (D-E) Part of PDF of the 40-year EQ hypocenters. Horizontal axis stands for EQ hypocenter location and vertical axis shows the PDF of the 12 directivities. (E) magnified view of some 40 hypocenters. (F) Sphere of potential rupture within the radius R . (G) When a large EQ occurs, the rupture length ($L_R \approx R$) can run in any direction within the sphere. Two distinct scenarios are shown.

TABLE 2. Evolution of potential energy function.

[Initialization]
Discretize domain V into reference volumes $\Delta V(\xi_j)$ with $(\Delta h, \Delta \lambda, \Delta \phi) = (5 \text{ km}, 0.1^\circ, 0.1^\circ)$
Initialize the maximum amplitude of PE $A_{max}(\xi_j) \leftarrow$ position-dependent (18)
Period of PE $P(\xi_j) \leftarrow 365000$ days (i.e., 1000 years), and
Phase of PE $\phi(t = 0.0; \xi_j) \leftarrow 0$ for $\forall j$
Loop t in $[1, n_{epoch}]$
[Task 1] Kinetic Energy Calculation
$E_K(t; \xi_j) = \sum_{\forall X_i \in V_j} 10^{(4.4+1.5 \times M_w(X_i))}$, $\forall \xi_j \in V$ and within one epoch where $M_w(X_i)$ is the observed moment magnitude at point X_i ; magnitude-energy rule of (5) is used
Loop for each reference volume $\xi_j \in V$
[Task 2] Rupture Range Calculation
$R(t, \xi_j)$ from $E_K(t; \xi_j)$ by (7)
Direction of $R(\xi_j) \leftarrow$ Rupture Direction Determination Algorithm (Fig. 4)
Define Ω_j of $R(\xi_j)$, the subdomain of the rupture zone.
[Task 3] Update Phase Shift of PE Function
Loop $i \in \Omega_j$
Calculate $T_K^{(dist)}(\xi_i)$ by (20). [i.e., distributed KE across Ω_j]
If $n(\Omega_j) = 1$ [i.e., small EQ confined within one reference volume]
$\Delta \phi_{small}$ from (10)
$\phi(t; \xi_i) \leftarrow \phi(t; \xi_i) + \Delta \phi_{small}$ [small phase shift, Fig. 3(A)]
If $n(\Omega_j) > 1$ [large EQ spans many reference volumes]
$\phi(t; \xi_i) \leftarrow \Delta \phi_{large}$ by (II-C) [i.e., Figs. 3(A)];
Adjust $\phi(t; \xi_i)$ s.t. $\phi(t; \xi_i) \in [-P(\xi_i), P(\xi_i)]$
[Task 4] Reset Period of PE Function
Loop $i \in \Omega_j$ [Ω_j from (11) or (12)]
If $n(\Omega_j) \geq 1$,
$P(\xi_i) = \min \left[(c_{decay})^{-\left(n(\Omega_j) + \text{remainder} \left(\frac{R(t, \xi_j)}{\max[\Delta X, \Delta Y, \Delta h]} \right)} \right) \times P_0, P_{past} \right]$
where $[\Delta X, \Delta Y, \Delta h]$ is the dimensions of one reference volume ΔV
$P_{past}(\xi_i) \leftarrow P(\xi_i)$.
[Task 5] Update PE
$V(t; \xi_j) = \text{Sawtooth}(t; A_{max}(\xi_j), P(\xi_j), \phi(t; \xi_j))$ from (8)
End Loop Each reference volume
End Loop All Epochs (i.e., time windows)

kinetic energy. It should be noted that this rupture directivity map is solely based on EQ catalog data, without requiring pre-defined fault maps. Importantly, this purely data-driven rupture directivity map is not a fixed one. It is a natural learning outcome and essentially subject to evolution by further training of the AI framework. Inclusion of well-known fault maps can be straightforward, which would be a constructive future extension.

E. REAL OBSERVATIONS TO KINETIC AND POTENTIAL ENERGY

We utilized the USGS earthquakes catalog [39] as our primary dataset. From the real observations of moment magnitude $M_w(X_i)$ at hypocenter X_i , we honor the moment-energy relation (5) to convert the observed M_w to energy with empirical coefficients $\alpha_1 = -2.93$ and $\beta_1 = 0.67$

$$E = 10^{\frac{1}{\beta_1} - \frac{\alpha_1}{\beta_1} M_w} \approx 10^{4.4+1.5 M_w}. \quad (13)$$

Then, we can calculate the kinetic energy E_K of every reference volume $V_j(\xi_j)$ (i.e., discretized volume of lithosphere

with dimensions of $5 \text{ km} \times 0.1^\circ \times 0.1^\circ$ [depth, longitude, latitude]) within one time window t as

$$E_K(t; \xi_j) = \sum_{\forall X_i \in V_j} 10^{(4.4+1.5 M_w(X_i))}. \quad (14)$$

Here, as long as a hypocenter falls into the range of a reference volume, i.e., $X_i \in V_j$, the associated energy is assigned to the reference volume's center ξ_j . The released kinetic energies of all reference volumes perturb the potential energy function of the surrounding domain, specifically affecting their phase ϕ , period P , and amplitude A_{max} . Following the update of these quantities, the new potential energy V at each reference volume at a time window t is calculated by (8). As implied in (8), the potential energy $V(t, \xi_j)$ at a given time window and a reference volume location is not a static value; rather, it constantly evolves as a function of ϕ , P , and A_{max}

$$\text{Potential Energy: } V = V(\phi(t, \xi_j), P(\xi_j), A_{max}(\xi_j)) \quad (15)$$

The detailed explanation of phase update is given in Eqs. (9) and (10). The following sections II-F and II-G detail the

update procedures for period and maximum amplitude of V , respectively. The entire procedures are summarized in Table 2.

F. LEARNING PERIOD OF POTENTIAL ENERGY FUNCTION

One of the central learning objectives is the period of PE function. It is assumed that all spatial locations can have different periods, and thus finding $P(\xi_j)$ for all reference volumes is another daunting large ill-posed problem. Previous research in geophysics offers invaluable clues. For instance, Cao and Liu [40] elucidates crustal deformation rate of the western U.S. by conducting full 3D simulations accounting for interaction between asthenospheric flow and lithospheric thickness step. The distribution of crust deformation rate offers hints at the period of PE function of this paper since the slope $A_{max}^*(\xi_j)/P(\xi_j)$ of the PE function is related to $\partial V/\partial t$ (marked by shaded triangle in Fig. 3(B)). From the definition of PE, we can infer that $\partial V/\partial t \propto \partial \gamma/\partial t$. Although we are uncertain about the quantitative value of every location's $P(\xi_j)$, we can initiate learning process by leveraging the basic knowledge (e.g., [40], [41]) of “the higher seismic activity, the shorter period of PE function.” Although we are inaccessible to the quantitative true period, previous research [40] suggests that the depth-average crustal deformation rate (i.e., the strain rate tensor's 2nd invariant) of western US approximately range in $(\sim 10^{-16}, \sim 10^{-14})$ [1/s]. Although this is not a direct indication, this paper starts with a power decay rule. Fig. 5 shows the impact of different decay coefficients of the period (denoted by $c_{decay} \in \mathbb{R}, c_{decay} > 1.0$). “Reset Period of PE Function” of Table 2 explains the period decay rule. For instance, when $c_{decay} = 2.511$ a sufficiently large EQ (e.g., rupture length span more than 4 reference volumes) may result in $2.511^{(4+1)} \approx 100$ times smaller period than that of intact region. Here, the power 4 is from the total involved reference volumes. This power decay rule is not fixed and will be updated by rule-learning ML. As shown in Fig. 5(A), a large decay coefficient, the period distribution becomes quickly saturated by the minimum period (here 365 days) whereas a small decay coefficient can help discern diverse period distributions (Fig. 5(B)).

G. MAXIMUM AMPLITUDE OF POTENTIAL ENERGY

One of the most intractable unknowns of the PE function (8) is the amplitude term $A_{max}(\xi_j)$ which physically means the maximum PE stored in the reference volume at ξ_j . If we assume that large EQs release all PE stored in reference volume, we can infer $A_{max}(\xi_j)$, which calls for in-depth investigation in the future extension of this paper. In this paper, all reference volumes are initially assigned with the default amplitude density of PE from the energy and rupture length relation (7) as

$$A_{max}^{(default)}(\xi_j) = \frac{10^{\frac{(\alpha_2 - \alpha_1)}{\beta_1}} R^{\frac{\beta_2}{\beta_1}}}{\Delta V_j} \quad (16)$$

The above equation with ΔV_j is written as

$$A_{max}^{(default)} = 10^{\frac{(\alpha_2 - \alpha_1)}{\beta_1}} R^{\frac{\beta_2}{\beta_1}} \left(4\pi \|\xi_j\|_2^2 \Delta h \frac{\Delta \lambda}{360^\circ} \frac{\Delta \phi}{180^\circ} \right)^{-1} \quad (17)$$

where coefficients $\alpha_1, \alpha_2, \beta_1$, and β_2 are from the energy and rupture length rule (7); the default rupture length of one reference volume's diagonal dimension $R = 11.1$ km; its average volume $\mathbb{E}_{V_j}[\Delta V_j] = 391.8 \text{ km}^3$ used. In view of the curved ellipsoidal lithosphere, the volume of the j th reference volume element ΔV_j [km³] is calculated by $\Delta V_j = 4\pi \|\xi_j\|_2^2 \Delta h \frac{\Delta \lambda}{360^\circ} \frac{\Delta \phi}{180^\circ}$ where h [km] is the depth, λ [°] is the longitude and ϕ [°] is the latitude. In all simulations, $(\Delta h, \Delta \lambda, \Delta \phi) = (5 \text{ km}, 0.1^\circ, 0.1^\circ)$. Considering the inherent heterogeneity in material and structures of lithosphere, it is plausible to admit that there exists substantial uncertainty in A_{max} . Assuming that we don't know the accurate distribution of A_{max} in the domain, the AI framework accommodate the variation of A_{max} as follows.

$$A_{max}(\xi_j) = A_{max}^{(default)} \times (1 + \sigma_{ES} \times s(\xi_j)), \quad s \sim \mathcal{N}(0, 1^2) \quad (18)$$

where $A_{max}^{(default)} = (R \times 10^{3.345})^3$ is the default A_{max} value defined in (16) and σ_{ES} is the global standard deviation used in ES algorithm in Table 3. During the global learning process of ES, A_{max} may become too small or too large which is unrealistic and leads to numerical errors. To prevent the issue, the upper and lower bounds of A_{max} are defined as $10^3 A_{max}^{(default)}$ and $10^{-3} A_{max}^{(default)}$, respectively.

H. GLOBAL RULE LEARNING BY EVOLUTION STRATEGIES (ES)

One of the central goals of the proposed framework is to learn hitherto unknown rules about seismogenesis (e.g., the PE's period, phase, amplitude) with the observed environments (here, EQ catalog over past 40 years). In view of the complexity of the multiple learning objectives and the absence of simple labeled training data sets, the class of reinforcement learning may be a promising candidate. Recently, the evolution strategies (ES) has proven an efficient alternative to the reinforcement learning class since it holds notable advantages – e.g., no need for backpropagation of gradients, independence of sparse or dense reward distributions, and clear scalability [42], [43], [44], [45]. For scalability, the advanced parallel algorithm of CPU grouping [46] is also adopted for this paper.

Let $F_{ES}(\cdot)$ be the stochastic return (i.e., an error measure between observed EQs and predictions by the best-so-far rules) and θ_{ES} be the hyperparameters of the rules about EQs. We lack access to derivatives of the environment (i.e., non-smoothness issue), which the ES will address. In essence, we adopt a Gaussian-blurred objective. θ_{ES} is assumed from an isotropic multivariate Gaussian with a fixed mean and covariance $\sigma^2 \mathbf{I}$. Thus, $\mathbb{E}_{\theta_{ES}} F_{ES}(\theta_{ES}) = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} F_{ES}(\theta_{ES} + \sigma \epsilon)$. Then, the score function estimator that is used for the

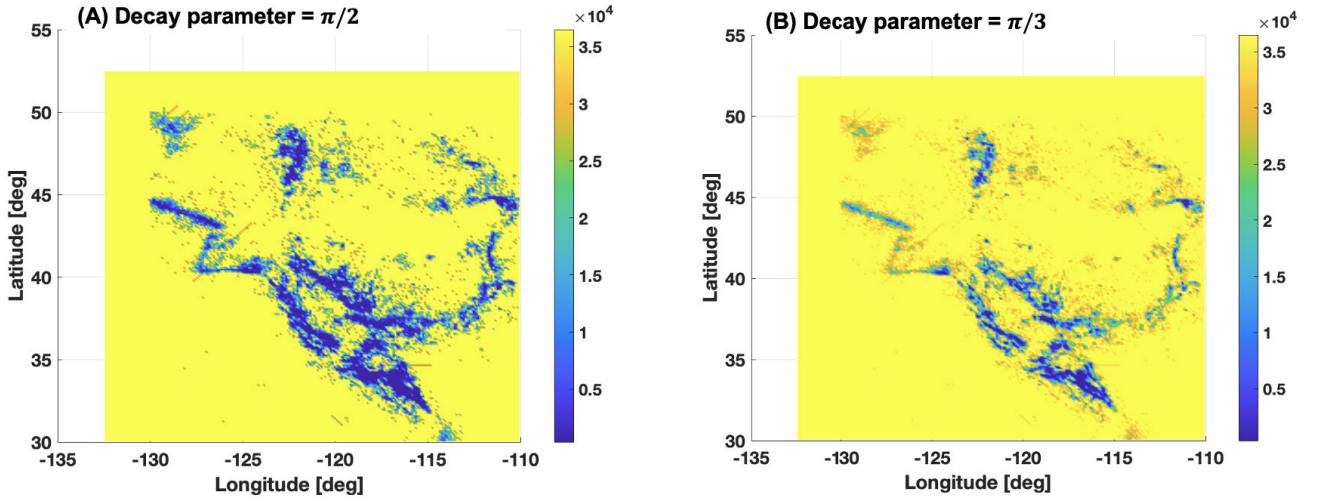


FIGURE 5. Period Distribution with Varying Decay Coefficient: Apparent impact of different decay coefficient. (A) the decay coefficient of the period is set as $\pi/2$ whereas (B) $\pi/3$. When a larger decay coefficient is used (A), the period distribution is quickly saturated by the minimum period (i.e., 365 days). Dark blue zones of (A) are relatively broader than (B).

TABLE 3. Global rule learning by parallel evolution strategies.

[Initialization]
α = the learning rate (default 0.1)
σ = the standard deviation of noise (default 0.1)
Prepare n_{wkr} workers with total available processors
n_{sample} = the total random samples
$\theta_{ES}^{(0)} \in \mathbb{R}^{n_{ES}}$ = inherit the prior-best hyperparameters of rules
Inherit the prior best period, phase, and amplitude of PE, $\{P, \phi, A_{max}\}$
Loop t in $[0, T_{ES}]$
Each Worker In Parallel: worker index $k \in [1, n_{wkr}]$
Distribute samples over workers: $n_{sample} = \sum_k n_{sample}^{(k)}$
Loop random sample $i = 1, \dots, n_{sample}^{(k)}$
Get noise $\epsilon_i^{(k)} \in \mathbb{R}^{n_{ES}}$; $\epsilon_i^{(k)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
$\bar{\theta}_{ES} = \theta_{ES}^{(t)} + \sigma \epsilon_i^{(k)}$
ECA: Training($\bar{\theta}_{ES}$) $\rightarrow$ updated $\{P, \phi, A_{max}\}^{(k)}$
ECA: Validation(frozen $\{P, \phi, A_{max}\}^{(k)}$) $\rightarrow F_{ES,i}^{(k)}$
End Loop
End of Each Worker
AllGather $F_{ES,i}^{(k)}$ from all workers
Stochastic Gradient Descent:
$\theta_{ES}^{(t+1)} = \theta_{ES}^{(t)} - \frac{\alpha}{\sigma \times n_{sample}} \sum_{i,k} F_{ES,i}^{(k)} \epsilon_i^{(k)}$
If new best performance is found, store θ_{ES} and $\{P, \phi, A_{max}\}$
End Loop

stochastic gradient descent is given by

$$\nabla_{\theta_{ES}} \mathbb{E}_{\epsilon} [F_{ES}(\theta_{ES} + \sigma \epsilon)] = \sigma^{-1} \mathbb{E}_{\epsilon} [\epsilon \times F_{ES}(\theta_{ES} + \sigma \epsilon)] \quad (19)$$

I. DISTRIBUTION OF KINETIC ENERGY

When an EQ is observed, this paper assumed that its energy is released via kinetic energy, denoted by $E_K(t; \xi_j)$, from hypocenter ξ_j . An intractable question is how we can distribute the released KE across lithosphere so that the affected reference volumes' potential energies are adjusted.

In general, the distributed kinetic energy is given as

$$T_K^{(dist)}(\xi_i) := \frac{\omega_i(\xi_j)}{\sum_{\forall i \in \Omega_j} \omega_i(\xi_j)} E_K(t; \xi_j) \quad (20)$$

where $\omega_i(\xi_j)$ determines the relative weight of i -th reference volume of the total KE originated from the hypocenter at j -th reference volume. This paper proposed and compared two approaches: (i) spherical distribution with distance-dependent Gaussian weight or (ii) line distribution with simple weighting of full/partial failure. In the first spherical distribution with distance-proportional weights, it is assumed that the reference volumes closer to the hypocenter are associated with the more KE with distance-disproportional Gaussian weight as

$$\omega_i(\xi_j) := (L(2\pi)^{1/2})^{-3} \exp\left(-\frac{|\xi_i - \xi_j|^2}{2L^2}\right) \quad (21)$$

Here, L is the spatial influence range, with which we can control the relative contribution of adjacent reference volumes to the released KE. In the second line distribution with simple weighting approach, the weights are determined as

$$\omega_i(\xi_j) = \mathcal{I}_R \times 1.0 + (1 - \mathcal{I}_R) \frac{R(\xi_i)}{\text{length}(V_i)} \quad (22)$$

where $\mathcal{I}_R$ is the indicator function that gives 0 when i -th reference volume is the end tip of the rupture and 1 otherwise; $R(\xi_i)$ is the rupture length at the i -th reference volume V_i . The end tip of the rupture means that the rupture does not fully occupy the dimension of the reference volume. Preliminary study found that the spherical distribution appears to over-smooth the energy across 3D space whereas the line distribution appears to better mimic the rupture distributions. The line distribution requires the rupture

directivity information which is also derived by past EQ data as explained in the main text.

III. PSEUDO-PROSPECTIVE PREDICTION RESULTS

A. PROCESSED EQ DATA FROM THE PAST 40-YEARS EQ CATALOG

This paper used the observed earthquake hypocenter data sets, USGS earthquakes catalog [39]. Details of EQ data processing are presented in the author's relevant works [17], [18], [19]. Briefly, the AI framework adopts the observed EQ hypocenter data sets which have been accumulated for the past 40 years. This AI framework process the raw data into a text-based matrix form of $\{\lambda, \phi, -h, M_w\}_i^{(t)}$, $i = 1, \dots, n^{(t)}$ and where $n^{(t)}$ means the number of total hypocenters recorded during one-time window in $[(t-1), t]$. The "time window" is a user-defined time range by which the continuous time space is discretized (details in [17] and [18]). In this paper, one time window corresponds to 30 days, and all AI-driven predictions of this paper mean the 30-day ahead EQ prediction with the prior EQs data. The coordinates $\{\lambda, \phi\}$ in [deg] are the longitude and latitude, respectively. h [km] is the ground-normal depth. For three-dimensional (3D) consideration of the AI framework, the geodetic coordinates $\{\lambda, \phi, -h, M\}_i^{(t)}$ are transformed into the earth-centered rectilinear coordinate $\{x, y, z, M_w\}_i^{(t)}$ [17], [18], [19]. All EQs of the past 40 years are processed, and the total number of EQ data points becomes 1,895,190. The day-based EQ data sets are shared for researchers (see [47]).

While the training of the AI framework used all the past 40-years EQ data, the pseudo-prospective prediction results presented in this paper are generated with large EQs $M_w \geq 7.0$. The list of the 8 large EQs is as follows.

- EQ1: 1991/8/17, Brookings EQ ($M_w = 7.0$),
- EQ2: 1992/4/25, Cape Mendocino EQ ($M_w = 7.2$),
- EQ3: 1994/9/1, Ferndale EQ ($M_w = 7.0$),
- EQ4: 1999/6/28, Landers EQ ($M_w = 7.3$),
- EQ5: 1999/10/16, Hector Mine EQ ($M_w = 7.1$),
- EQ6: 2005/6/15, Big Lagoon EQ ($M_w = 7.2$),
- EQ7: 2010/4/4, El Mayor-Cucapah EQ ($M_w = 7.0$),
- EQ8: 2019/7/6, Ridgecrest EQ ($M_w = 7.1$).

B. PREDICTION RESULTS AND COMPARISON TO UCERF3-ETAS

The target large EQs are eight EQs of the observed M_w being greater or equal to 7.0 in the western U.S. region since year 1980. The proposed LAP was pre-trained with all EQs from year 1980 through 2020 whereas the up-to-date UCERF3-ETAS as of year 2024 was used these predictions. All predictions are about the future EQs 30 days before the actual recorded events. It is called "pseudo-prospective" prediction since the pre-trained rules or present version UCERF3-ETAS predict 30 days before the EQs, instead of truly future predictions. For instance, Fig. 6 and Fig. 7 show pseudo-prospective predictions by the best-so-far AI-driven

rules that incorporate the LAP-based features. Although the AI-predicted EQs contain some false alarms (i.e., incorrect large EQs at wrong locations), the real large EQs are closely predicted by the LAP-based rules. Details of the prediction steps are summarized in Table 4.

To provide a relative performance of this LAP-based prediction, this paper obtained similar future predictions by using one of the well-established EQ forecasting frameworks, "UCERF3-ETAS" [25], [26], [48], [49]. As shown in Fig. 6 and Fig. 7, the precise prediction of rare large EQ is an intractable task. The use of UCERF3-EAS may not be perfectly suitable, since EQ forecasting's goal is not for short-term deterministic predictions of specific large EQs. However, this comparative investigation offers a relative standing of this paper in comparison to existing EQ forecasting approaches. Detailed settings for UCERF3-ETAS are presented in the relevant work of the author [19].

For the detailed comparison, Fig. 9 summarizes the pseudo-prospective predictions of the pre-trained LAP and the up-to-date UCERF3-ETAS. As shown in the third column (header = M_w), the best-so-far LAP's prediction of magnitude is deterministic while that of UCERF3-ETAS is probabilistic (i.e., $\log_{10}(M \geq M_{threshold}) \approx -3.0$). Depending upon the target EQ, $M_{threshold}$ varies from 4.0 to 6.0 (see M_w column of Fig. 9). UCERF3-ETAS predictions of large EQ events generally exhibit the low probability of the occurrence, e.g., $\log_{10}(Prob[M \geq 6.0]) \approx -3.0 \rightarrow Prob[M \geq 6.0] \approx 0.001$ whereas the errors in peak magnitudes predicted by the best-so-far LAP range from 0.0 to 0.5 (see the column of ΔM_w). Although the magnitude prediction of UCERF3-ETAS shows low probability (i.e., in the order of 0.1%), its prediction accuracy of hypocenter location is noteworthy. The last four columns of Fig. 9 present the predicted hypocenter of EQ, i.e., λ, ϕ, h and their error $\Delta\xi$ [km]. Overall, the best-so-far LAP's prediction accuracy of the hypocenter appears comparable to that of the up-to-date UCERF3-ETAS (see the last column of Fig. 9).

Fig. 10 summarizes the quantitative error metrics between the observed data and two pseudo-prediction methods. The proposed LAP-based model outperforms UCERF-ETAS in predicting 30-day maximum magnitudes, though both demonstrate comparable accuracy in hypocenter location. One difference is that UCERF-ETAS does not provide the depth h prediction of hypocenter, while the LAP-based prediction provides three-dimensional hypocenter prediction (last column of Fig. 10).

Importantly, the proposed LAP-based predictions are made possible by thorough pre-training with the entire EQs in the past 40 years. The outcome of the training takes not a black-box style fine tuning but a transparent rules. The training process continues improving the underlying PE function's characteristics (e.g., period and phase in Fig. 8) at each reference volume. For instance, the best-so-far PE functions are uniquely defined at each reference volume, and Fig. 8 shows distributions of decisive terms of period and phase

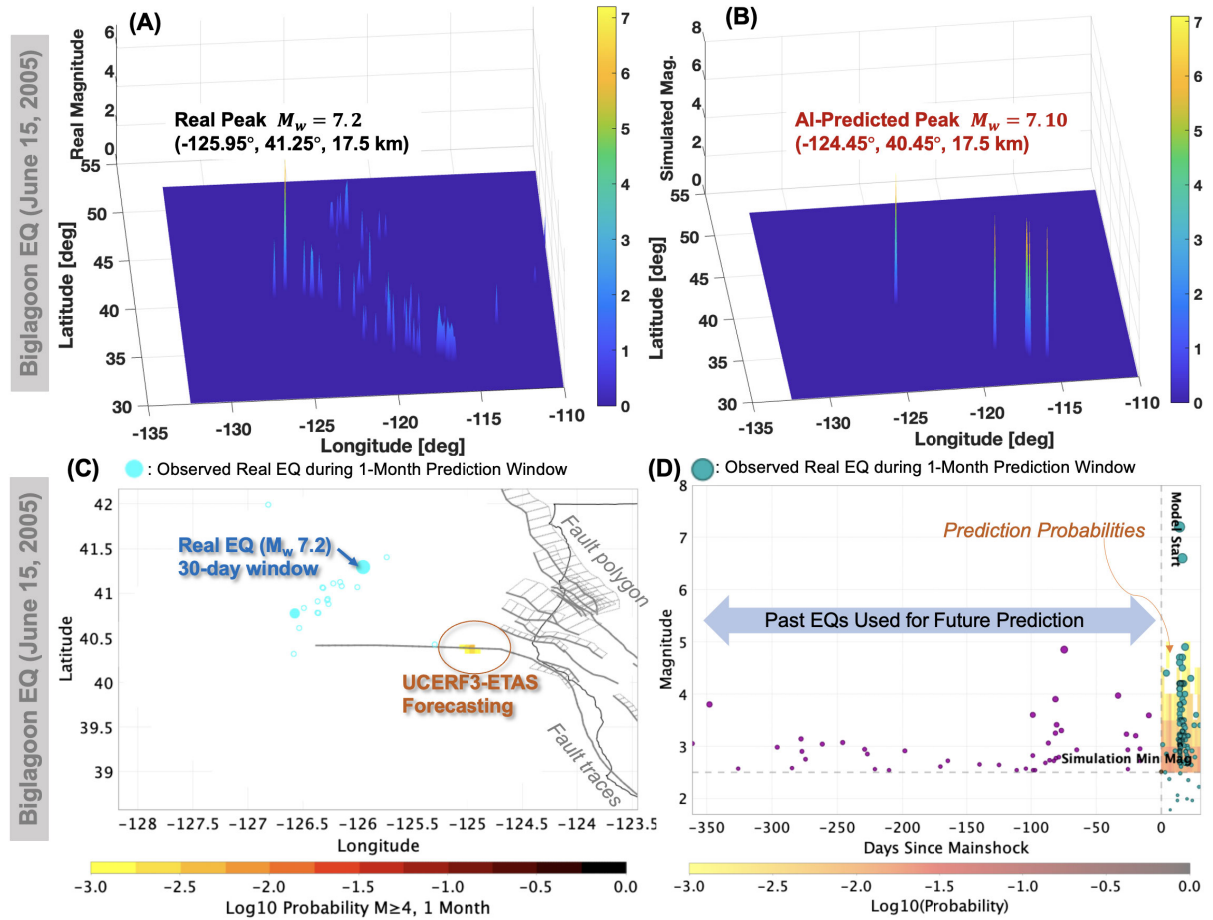


FIGURE 6. Pseudo-Prospective Prediction 30 days before the Big Lagoon EQ: (A) Real Observed EQs; (B) LAP-based predictions with the frozen PE and rupture directivity. UCERF3-ETAS Forecastings 30 days before the EQ: (C) ComCat probability spatial distribution from UCERF3-ETAS; (D) magnitude versus time probability function (adapted from [19]).

TABLE 4. Evolution of EQ generation rules.

[Initialization]	
Frozen the rupture directivity and the best-so-far PE:	
$A_{max}(\xi_j)$ the maximum amplitude, $P(\xi_j)$ the period, and $\phi(\xi_j)$ the phase of PE.	
Loop t in $[1, n_{epoch}]$	
Loop for each $\xi_j \in V$	
[Task 1] Calculate PE	
$V(t; \xi_j) = \text{Sawtooth}(t; A_{max}(\xi_j), P(\xi_j), \phi(\xi_j))$ from (8)	
[Task 2] Calculate LAP-based feature	
$6\ddot{R}\ddot{R}(t; \xi_j)$ from (30) and (31)	
[Task 3] Find ridges with the connected peaks of the LAP-driven features	
Recursive search for the connected peaks of $6\ddot{R}\ddot{R}$	
A Rupture Ridge $R_R := \{\dots, \xi_{(i)}, \xi_{(i+1)}, \dots\}$ s.t. $ 6\ddot{R}\ddot{R} > 2 \times 10^7$	
and $\ \xi_{(i)} - \xi_{(i+1)}\ \leq R$	
Append the new rupture ridge to the total ridge set $\mathcal{R}_R, R_R \rightarrow \mathcal{R}_R$.	
[Task 4] Predict the moment magnitude (M_{pred}) distribution	
For each rupture ridge $R_{Ri} \in \mathcal{R}_R$, use a logarithmic rule similar to (6)	
$M_{pred}(\xi_c) \leftarrow c_1^{(LAP)} \log(R_{Ri}) + c_2^{(LAP)}$ where ξ_c is rupture ridge center	
[Task 5] Error calculation with the observed M_w and predicted M_{pred}	
$Err(t) \leftarrow \mathcal{J}(M_w, M_{pred})$:	
Errors in Magnitude, Distance, and False Alarms (33)	
End Loop Each reference volume	
End Loop All Epochs (i.e., time windows)	
$F_{ES} = \mathbb{E}_t[Err(t)]$	

of the PE functions along with LAP-based feature $6\ddot{R}\ddot{R}$ and rupture directivity. All of the characteristics of PE function

at individual reference volumes and the EQ directivity are learned outcomes and helpful for interpreting seismogenesis

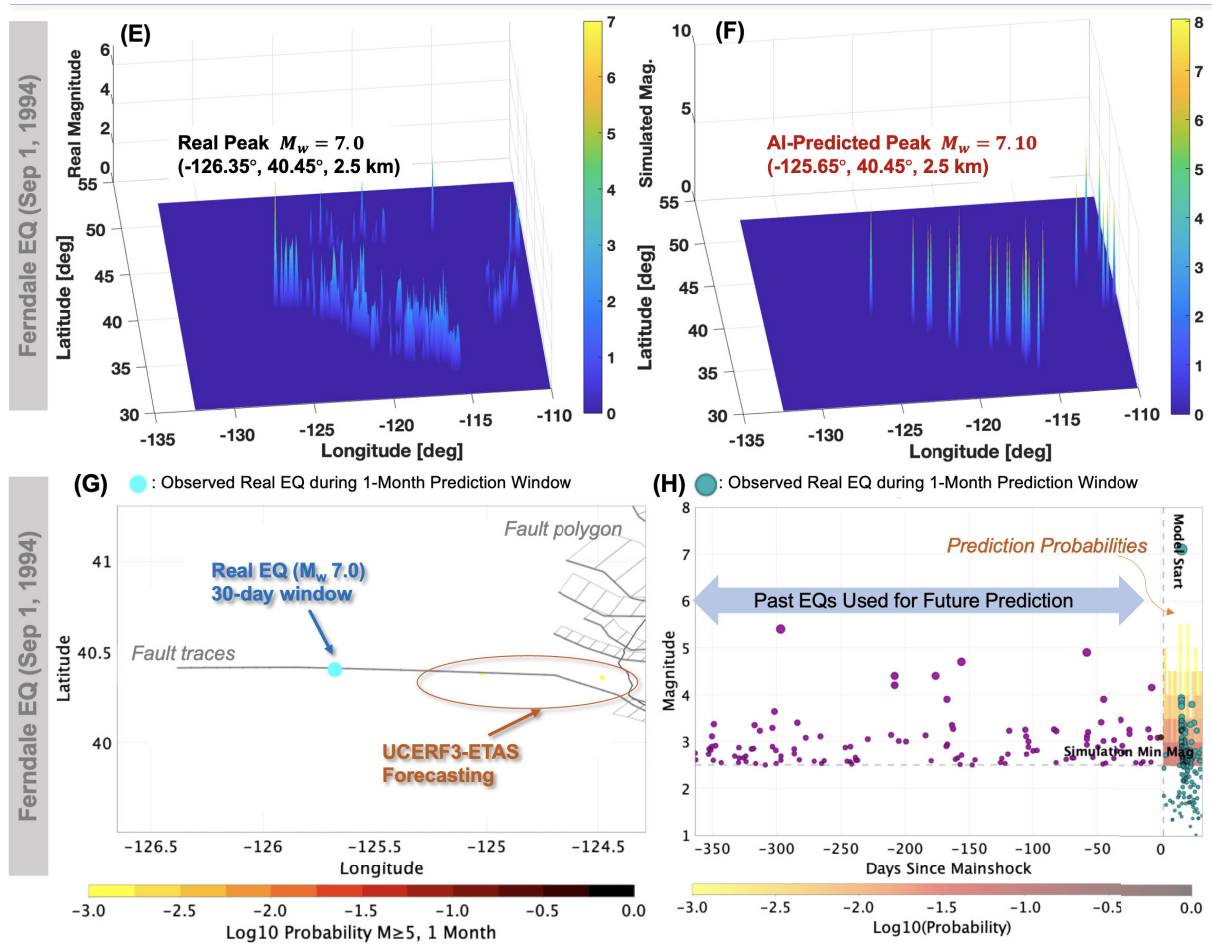


FIGURE 7. Pseudo-Prospective Prediction 30 days before the Ferndale EQ: (A) Real Observed EQs; (B) LAP-based predictions with the frozen PE and rupture directivity. UCERF3-ETAS Forecastings 30 days before the EQ: (C) ComCat probability spatial distribution from UCERF3-ETAS; (D) magnitude versus time probability function (adapted from [19]).

nature and for improving the EQ generation models in the future extensions with additional domain-specific expertise or deeper knowledge in geophysics and seismology. The evolution is naturally open for the proposed approach. Such a transparent leaning outcome from the proposed AI training is a notable difference from black-box style ML methods.

For the interested researchers, the LAP-based predictions of all eight large EQs of the past 40 years in the western U.S. region are presented in Fig. 11 and in Fig. 13. Also, Fig. 12 and Fig. 14 presents all UCERF3-EAS predictions for all eight large EQs.

Fig. 11 shows the LAB-based pseudo-prospective predictions of large EQs 30 days in advance between 1999 and 2019. When compared to the observed real EQs (shown in left column), the location and the maximum magnitude of the hypocenter is promising. Although some false alarms occur—indicated by incorrect peaks in the right column—the LAB-based AI framework successfully reproduces large events after training on 40 years of seismic data. Similar performance can be found in Fig. 13, showing the LAB-based pseudo-prospective predictions of large EQs within

1991 through 1994. In contrast to the LAP-based predictions, UCERF3-ETAS provides probabilistic forecasts of the locations of large EQs with a promising accuracy (marked by circle in Figs. 12 and 14). But the consistently low probabilities of the large EQs of next 30 days (indicated by $\text{Log}_{10}[\text{Probability}]$ of each panel) appear to underscore the limitation of UCERF3-ETAS for short-term prediction of large EQ.

IV. DISCUSSION

A. FORM OF KINETIC ENERGY

This paper assumes that the kinetic energy of EQ is in a form of rupture velocity $\dot{R}$, but its explicit form is unknown. If we take a simple linear form $T = c\dot{R}$, then $\partial T / \partial \dot{R} = c$, constant, leaving $d(c)/dt = -\partial V / \partial R = 0$. This leads to a trivial conclusion. If we take a quadratic form $T = c(\dot{R})^2$, then $\partial T / \partial \dot{R} = 2c\dot{R}$, leading to $d(2c\dot{R})/dt = 2c\ddot{R} = -\partial V / \partial R$. This implies a monotonic (probably unrealistic) behavior of rupture velocity. In this paper, as illustrated in Fig. 3(B), we assume that when EQ occurs (i.e., $R(t) > 0$), the PE decreases, i.e., $\partial V / \partial R < 0$. This means $2c\ddot{R} > 0$ and thus

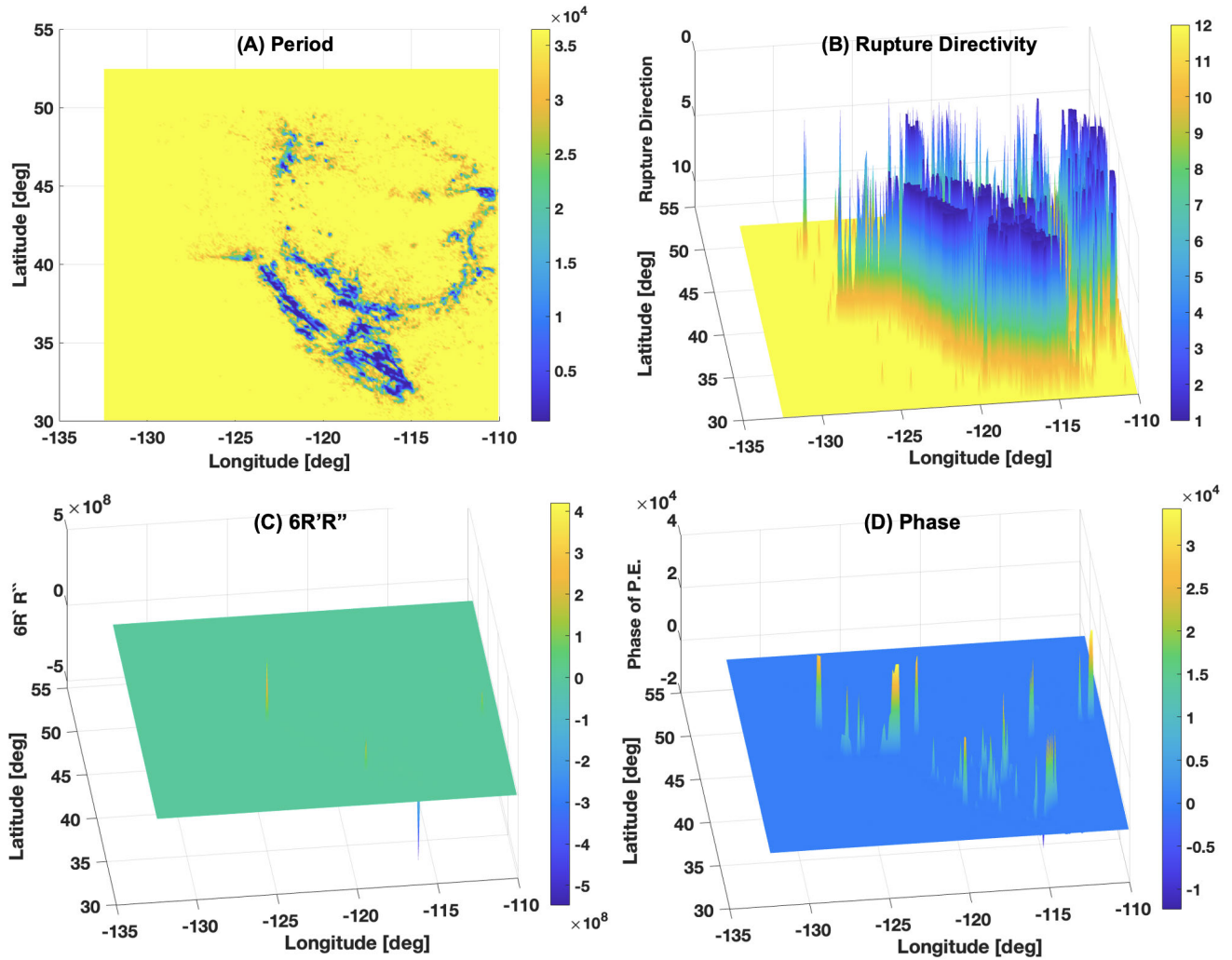


FIGURE 8. Distributions of the Best-So-Far PE Functions at Depth 7.5 km Used for the Pseudo-Prospective Predictions. (A) Period Distribution: Starting from 36500 days, the period of PE function is evolved with 40 years EQ data by using a period decay rule. (B) Rupture Directivity Distribution: By using the observed EQs, rupture directivity keeps evolving. Each color means different rupture direction. (C) LAP-Driven Feature $6R'R''$: At the last time step, the plot shows a snapshot of distribution of $6R'R''$ which keeps changing with time and EQ events. (D) Phase Distribution: Phase of the PE may undergo small shift, large shift, or no shift subject to small EQ, large EQ, or aseismic stage, respectively.

$\ddot{R} > 0 \rightarrow \dot{R}(t_2) > \dot{R}(t_1), \forall t_2 > t_1$. Physically, this may mean the rupture velocity keeps increasing monotonically, regardless of the size of EQ magnitudes. This monotonic behavior in $\dot{R}$ can be found in the higher-order forms too. If we take a fourth-power form $T = c(\dot{R})^4$, then $\partial T / \partial \dot{R} = 4c\dot{R}^3$, leading to $d(4c\dot{R}^3)/dt = 12c\dot{R}^2\ddot{R} = -\partial V / \partial R$. When EQ occurs, i.e., $\partial V / \partial R < 0$, we again have $12c\dot{R}^2\ddot{R} > 0$ and thus $\ddot{R} > 0 \rightarrow \dot{R}(t_2) > \dot{R}(t_1), \forall t_2 > t_1$. This is also a monotonic increasing behavior which is probably unrealistic.

However, if $T(\dot{R}) = c\dot{R}^3$, we can obtain physically and mathematically meaningful behaviors.

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{R}_j} \right) = \frac{d}{dt} (3c\dot{R}_j^2) = 6c\dot{R}_j\ddot{R}_j = -\frac{\partial V}{\partial R_j}. \quad (23)$$

This implies that $\text{sign}(\dot{R})$ must be identical to $\text{sign}(\ddot{R})$ whenever EQ occurs, i.e., $\partial V / \partial R < 0$. This provides a

bounded condition unlike other cases, and more importantly, it exhibits “cyclic” behaviors. In other words, a rupture develops only when the sign of $\ddot{R}\dot{R}$ become in align with that of $\partial V / \partial R$; a rupture ceases otherwise. It should be noted that we obtain this “cyclic” nature although we have not yet intentionally introduced any periodic nature to the $\partial V / \partial R$ nor to V .

B. LIMITATIONS

There are limitations that call for future extensions. This paper did not explicitly pay attention to the non-conservative forces (e.g., frictions or dissipations) when adopting the Hamiltonian approach and deriving the Euler-Lagrange equations. However, the role of frictions and energy dissipation is one of the central mechanisms of modulating and quenching EQs [24]. This paper focuses on the onset of large EQs and elucidates a potential role of the least action principle

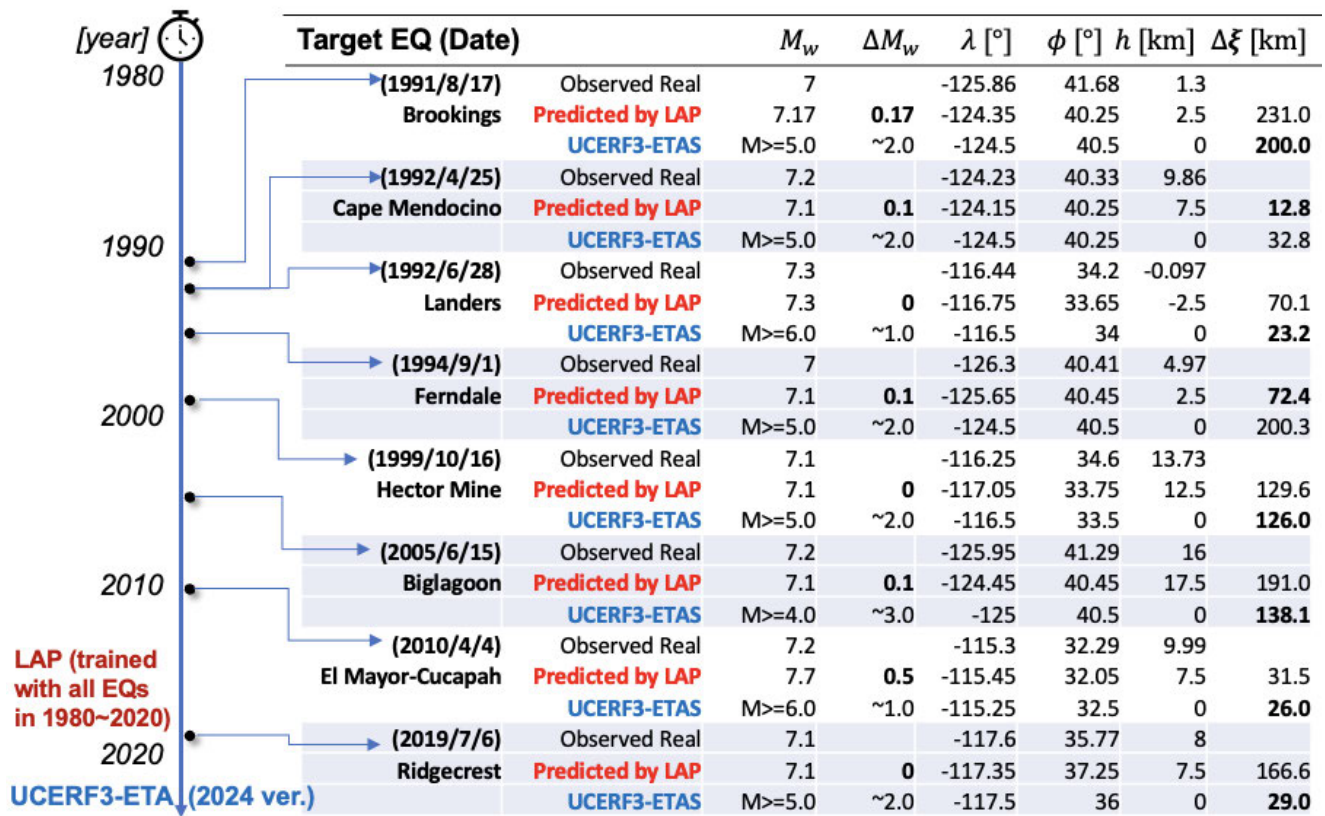


FIGURE 9. Summary of 30 days-Ahead Pseudo-Prospective Predictions of the LAP and UCERF3-ETAS: LAP pre-trained with all EQs between years 1980 and 2020 was used to predict the eight large EQs ($M_w \geq 7.0$) since year 1980 in the western U.S. region. The up-to-date UCERF3-ETAS as of year 2024 was used for similar predictions.

from the Hamiltonian mechanics perspective, which offers a meaningful conclusion. Also, in this paper, the PE function intentionally takes a sawtooth shape and thus the maximum possible PE is naturally limited.

V. CONCLUSION

According to the well-established consensus of the research community [16], deterministic predictions of EQs remain impossible, which this paper essentially agrees with. But, persistent efforts to understand large EQs are of paramount importance. To add a meaningful new dimension to the endeavor, this paper presents a novel perspective that regards the EQ-bearing lithosphere as a chaotic system and elucidates whether the least action principle of Hamiltonian Mechanics can shed light on the onset of large EQs. From the abstract space of time and rupture derivatives, this paper's investigation demonstrates a potential causal pathway between LAP and large EQs. For the computational investigation, this paper developed an evolving ML of parallel evolution strategies to help unravel the hidden rules based on LAP. After training with the entire EQs of the past 40 years in the western U.S. region, this paper's feasibility test results with the historic large EQs appear promising and adds a new dimension to the pursuit of large EQ. The findings of this paper can be incorporated into advanced AI approaches, e.g.,

reinforcement learning for EQs [19], deep learning methods for EQ data [10], and sophisticated EQ prediction rules that use other AI-friendly features of EQs [17], [18]. The feasibility test results for the past 40 years of large EQs in the western U.S. region appear promising compared to state-of-the-art EQ forecasting methods.

The proposed framework utilizes a one-month "prediction horizon," established by a consistent 30-day time window throughout the training and feasibility tests. However, this horizon is inherently flexible; it can be easily adjusted to weeks or days simply by modifying the time window. For example, a one-week ahead prediction would only require shifting the window to 7 days for both rule-learning and prediction (i.e., inference).

Although this paper's conclusion is far from practical applicability and calls for deeper investigations, the result shows a promising new research direction to large EQs based on a fusion of classical physics and evolving ML.

Future Work: Substantial future extensions are required to address the aforementioned limitations of this study. For instance, future extensions should incorporate the detailed non-conservative term (e.g., friction, dissipation, etc.) in the right-hand side of the Euler-Lagrangian equations. Since the precise description of friction or dissipation across a broad region inside the lithosphere may raise another formidable

Target EQ		M_w	$MAE[M_w]$	λ	$MAE[\lambda]$	ϕ	$MAE[\phi]$	h	$ \Delta h [\text{km}]$
(1991/8/17) Brookings	Observed Real	7		-125.86		41.68		1.3	
	Predicted by LAP	7.17	2.4%	-124.35	1.2%	40.25	3.4%	2.5	1.2
	UCERF3-ETAS	$M \geq 5.0$	30.3%	-124.5	1.1%	40.5	2.8%	0	N/A
(1992/4/25) Cape Mendocino	Observed Real	7.2		-124.23		40.33		9.86	
	Predicted by LAP	7.1	1.4%	-124.15	0.1%	40.25	0.2%	7.5	2.4
	UCERF3-ETAS	$M \geq 5.0$	29.6%	-124.5	0.2%	40.25	0.2%	0	N/A
(1992/6/28) Landers	Observed Real	7.3		-116.44		34.2		-0.097	
	Predicted by LAP	7.3	0.0%	-116.75	0.3%	33.65	1.6%	-2.5	2.4
	UCERF3-ETAS	$M \geq 6.0$	17.8%	-116.5	0.1%	34	0.6%	0	N/A
(1994/9/1) Ferndale	Observed Real	7		-126.3		40.41		4.97	
	Predicted by LAP	7.1	1.4%	-125.65	0.5%	40.45	0.1%	2.5	2.5
	UCERF3-ETAS	$M \geq 5.0$	29.6%	-124.5	1.4%	40.5	0.2%	0	N/A
(1999/10/16) Hector Mine	Observed Real	7.1		-116.25		34.6		13.73	
	Predicted by LAP	7.1	0.0%	-117.05	0.7%	33.75	2.5%	12.5	1.2
	UCERF3-ETAS	$M \geq 5.0$	29.6%	-116.5	0.2%	33.5	3.2%	0	N/A
(2005/6/15) Biglagoon	Observed Real	7.2		-125.95		41.29		16	
	Predicted by LAP	7.1	1.4%	-124.45	1.2%	40.45	2.0%	17.5	1.5
	UCERF3-ETAS	$M \geq 4.0$	43.7%	-125	0.8%	40.5	1.9%	0	N/A
(2010/4/4) El Mayor-Cucapah	Observed Real	7.2		-115.3		32.29		9.99	
	Predicted by LAP	7.7	6.9%	-115.45	0.1%	32.05	0.7%	7.5	2.5
	UCERF3-ETAS	$M \geq 6.0$	22.1%	-115.25	0.0%	32.5	0.7%	0	N/A
(2019/7/6) Ridgecrest	Observed Real	7.1		-117.6		35.77		8	
	Predicted by LAP	7.1	0.0%	-117.35	0.2%	37.25	4.1%	7.5	0.5
	UCERF3-ETAS	$M \geq 5.0$	29.6%	-117.5	0.1%	36	0.6%	0	N/A

FIGURE 10. Errors of Pseudo-Prospective Predictions of the LAP and UCERF3-ETAS: Mean absolute error (MAE) between quantities of the observed hypocenter and predicted values of Fig. 9. UCERF3-ETAS does not predict quantity at a three-dimensional depth, and thus h is marked by 'N/A'.

challenge. Thus, future extensions should conduct in-depth investigations and involve diverse researchers in collaborative efforts to elucidate the effects of friction and dissipation on the proposed framework of this paper. As a long-term research topic, statistical and risk analyses of the entire EQs should be conducted using the proposed framework. And the resultant conclusions or trends should be compared with other existing statistical and risk-oriented methods.

APPENDIX A

A. DERIVATION OF RUPTURE DERIVATIVES FROM LAP

The generic solution to the LAP in the system of n generalized coordinates (i.e., rupture zones) is the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial(T-V)}{\partial \dot{R}_j} \right) - \frac{\partial(T-V)}{\partial R_j} = 0, \quad (j = 1, \dots, n) \quad (24)$$

Since we assume $T = T(\dot{R})$ and $V = V(R)$, (2) is simplified as

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{R}_j} \right) = - \frac{\partial V}{\partial R_j}, \quad (j = 1, \dots, n) \quad (25)$$

As explained in Discussion section, this paper has $T(\dot{R}) = c(\dot{R})^3$ with c , a constant. Focusing on the right-hand side of the (23), this paper assumes that the PE follows a sawtooth function $V = V(R(t), t)$ of (8). Thus, we can have by the chain rule

$$- \frac{\partial V}{\partial R_j} = - \frac{\partial V}{\partial t} \frac{\partial t}{\partial R_j} \quad (26)$$

$$- \frac{\partial V}{\partial R_j} = \frac{2A_{max}}{P} \sum_{k=1}^{n_{st}} (-1)^k \cos \left(\frac{2\pi k}{P} (t + \phi) \right) \left(\frac{\partial R_j}{\partial t} \right)^{-1} \quad (27)$$

where the location-dependent terms $A_{max}(\xi)$, $P(\xi)$ and $\phi(t, \xi)$ are assumed to be the best-so-far fixed values after the prior training process (e.g., Table 2). Therefore, the Euler-Lagrange equations to the LAP for EQ generation boil down to the following equations

$$\ddot{R}_j^2 \ddot{R}_j = \frac{A_{max}}{3cP} \sum_{k=1}^{n_{st}} (-1)^k \cos \left(\frac{2\pi k}{P} (t + \phi) \right). \quad (28)$$

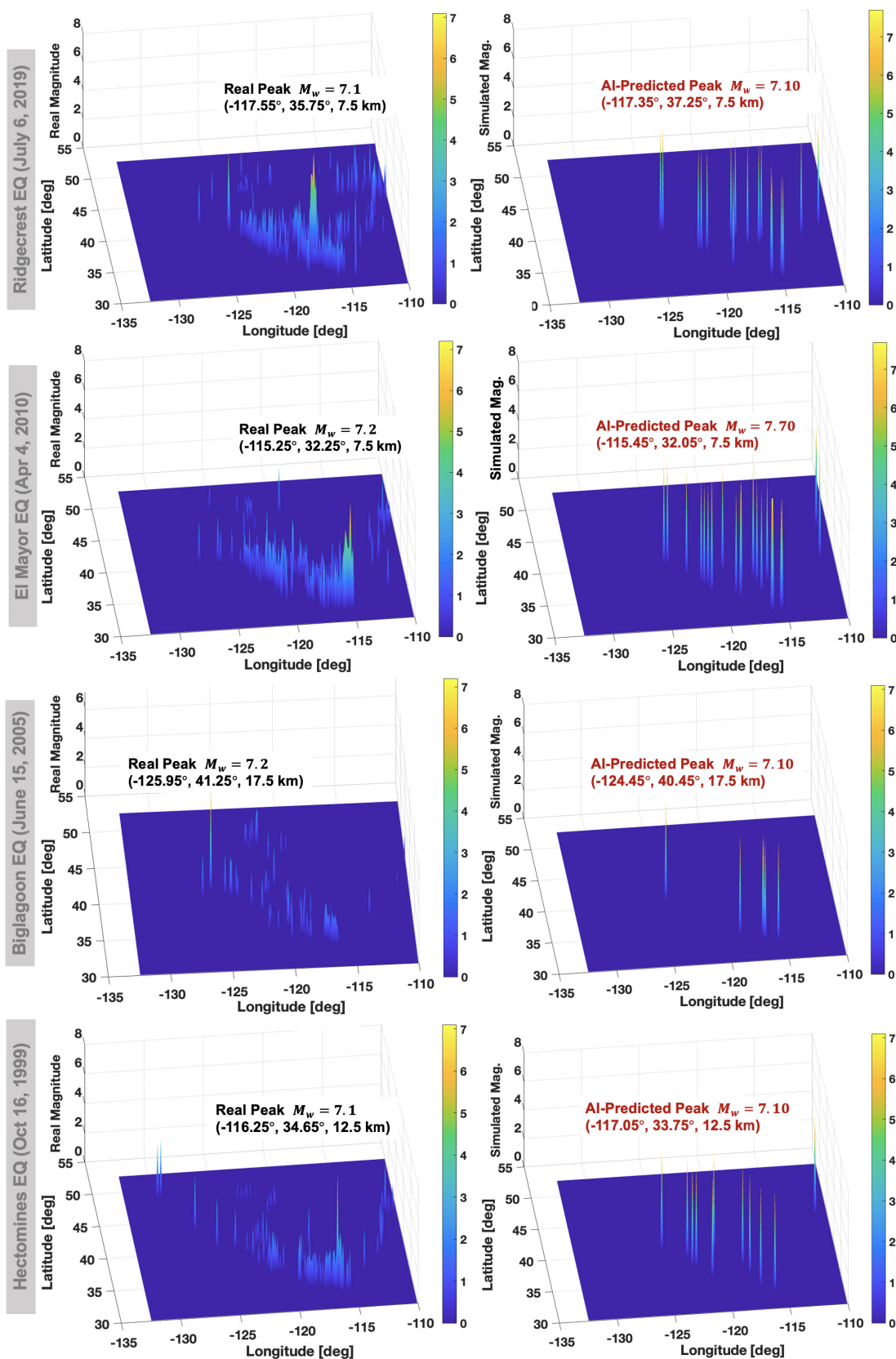


FIGURE 11. Pseudo-Prospective Prediction 30 days before Real EQs: (Left Column) Real Observed EQs; (Right Column) LAP-based predictions with the frozen potential energy and rupture directivity.

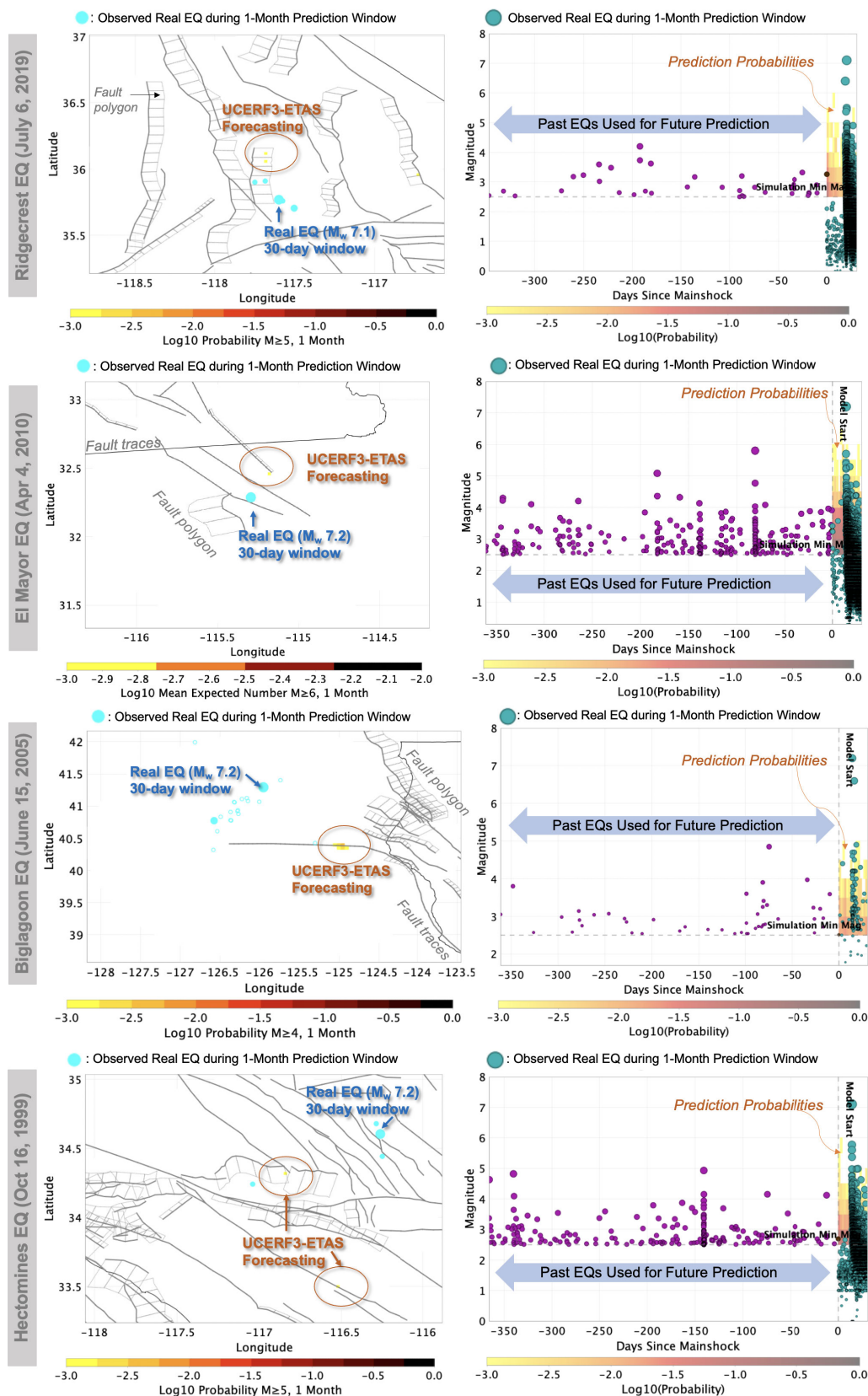


FIGURE 12. UCERF3-ETAS Forecastings 30 days before Real EQs: (Left Column) ComCat probability spatial distribution from UCERF3-ETAS; (Right Column) magnitude versus time probability function (adapted from [19]).

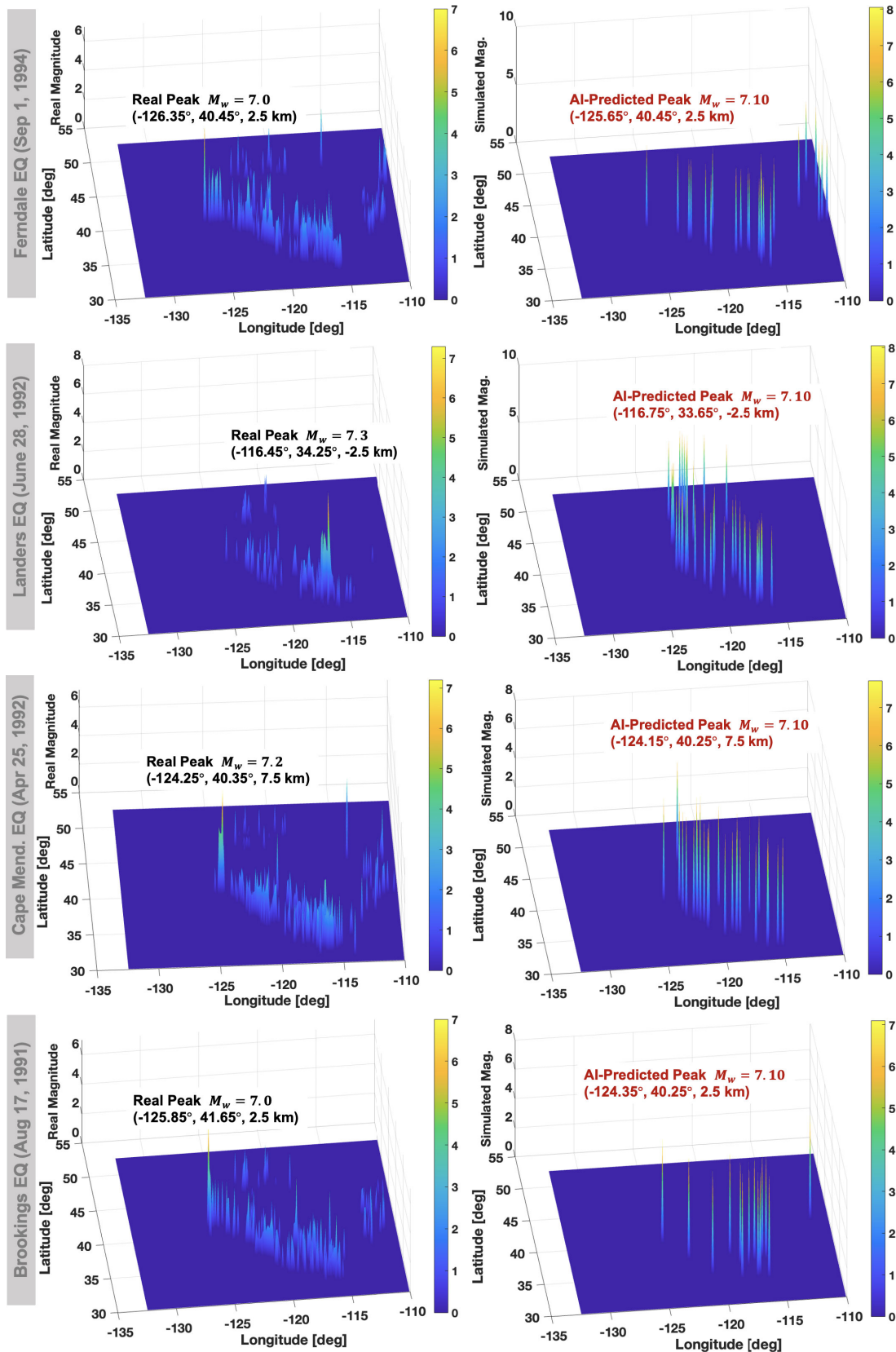


FIGURE 13. Pseudo-Prospective Prediction 30 days before Real EQs: (Left Column) Real Observed EQs; (Right Column) LAP-based predictions with the frozen potential energy and rupture directivity.

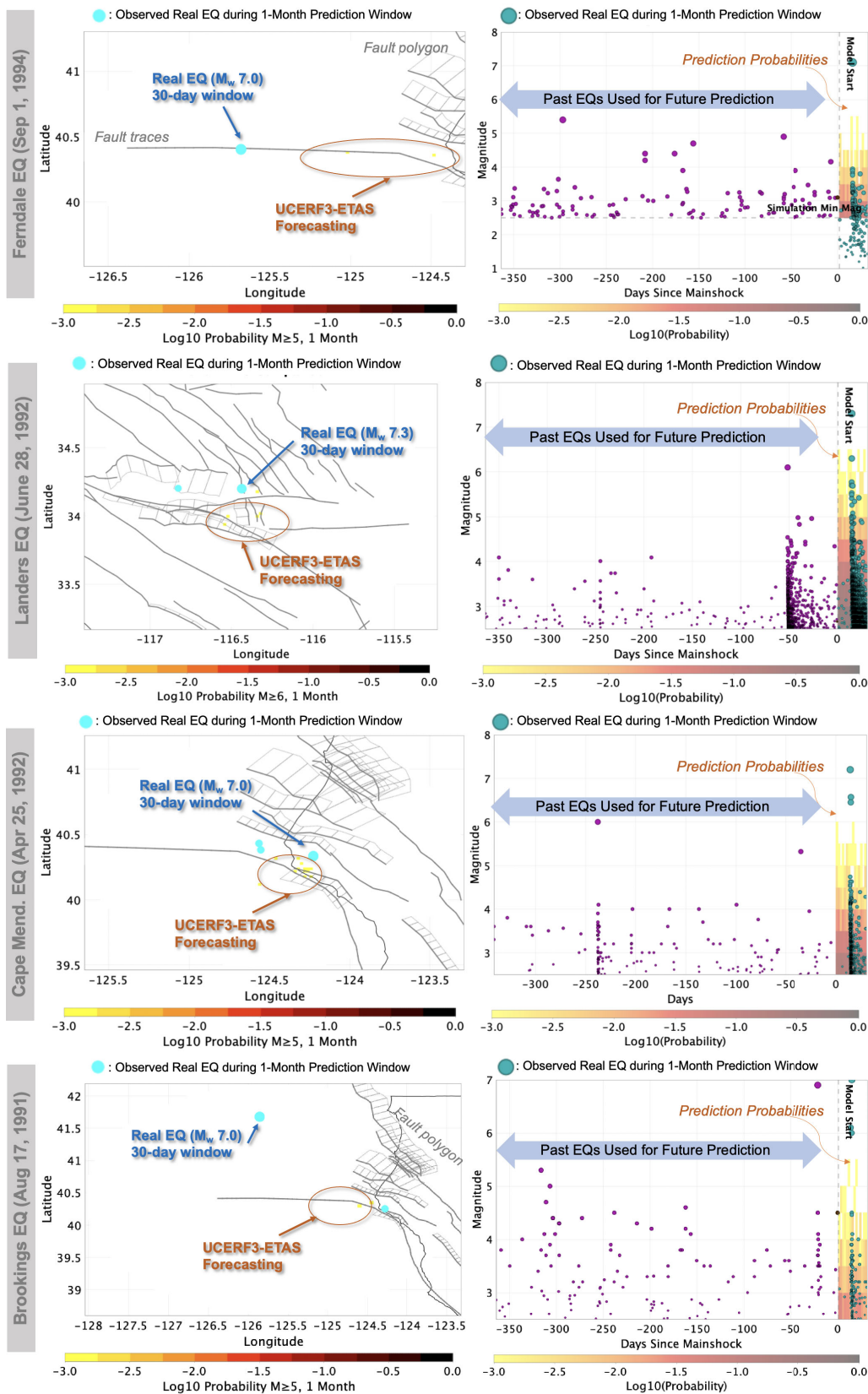


FIGURE 14. UCERF3-ETAS Forecastings 30 days before Real EQs: (Left Column) ComCat probability spatial distribution from UCERF3-ETAS; (Right Column) magnitude versus time probability function (adapted from [19]).

By solving these partial differential equations, we can obtain the LAP-driven EQ rupture rule as

$$R_j(t) = \int \left[\frac{A_{max}}{2\pi c} \sum_{k=1}^{n_{st}} \frac{(-1)^k}{k} \sin \left(\frac{2\pi k}{P} (t + \phi) \right) + C \right]^{1/3} dt, \quad (29)$$

where c is the coefficient of $T = c\dot{R}^3$ and C is a constant from the integration process. The hypocenter of the j th rupture is assumed to be related to the location ξ_j and thus $A_{max}(\xi)$, $P(\xi)$ and $\phi(t, \xi)$ can be related to the best-so-far values therein.

$$\dot{R}_j(t) = \left[\frac{A_{max}}{2\pi c} \sum_{k=1}^{n_{st}} \frac{(-1)^k}{k} \sin \left(\frac{2\pi k}{P} (t + \phi) \right) + C \right]^{1/3}. \quad (30)$$

$$\ddot{R}_j(t) = \frac{A_{max}}{3cP} \dot{R}_j^{-2} \sum_{k=1}^{n_{st}} (-1)^k \cos \left(\frac{2\pi k}{P} (t + \phi) \right). \quad (31)$$

These two equations are inserted into the following relation (from (23)) that can offer direct relationship between the rupture dynamics and PE variation:

$$\frac{\partial V}{\partial R_j} = -6c\dot{R}_j\ddot{R}_j \quad (32)$$

B. THREE-FOLD ERROR RULE OF MAGNITUDE, DISTANCE, AND FALSE ALARMS

This paper adopted the three-fold error rule that comprehensively incorporates discrepancies in magnitude and distance as well as the false alarms. The key error equation is presented below and the full details are given in [17] and [18].

$$\mathcal{J} = (1 - a_{cnt}) \sum_{\tilde{k} \in Top} \omega_{MD}^{(\tilde{k})} E_{MD}^{(\tilde{k})} / n(Top) + a_{cnt} E_{cnt} \quad (33)$$

where $E_{MD}^{(\tilde{k})}$ is the error in magnitude and 3D spatial distance of the real and predicted hypocenters. E_{cnt} is the error in the total count of false alarms. a_{cnt} is the coefficient about the relative importance between $E_{MD}^{(\tilde{k})}$ and E_{cnt} (0.1 is used). Top means the set of indices of reference volumes that contains the sorted real magnitudes ≥ 6.5 . $\omega_{MD}^{(\tilde{k})} := \exp(M_{obs}^{(t+1)}(\xi_{\tilde{k}})/10.0)$ meaning the magnitude-dependent scale-up factor.

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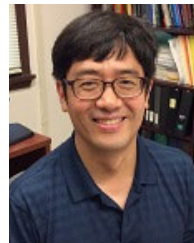
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